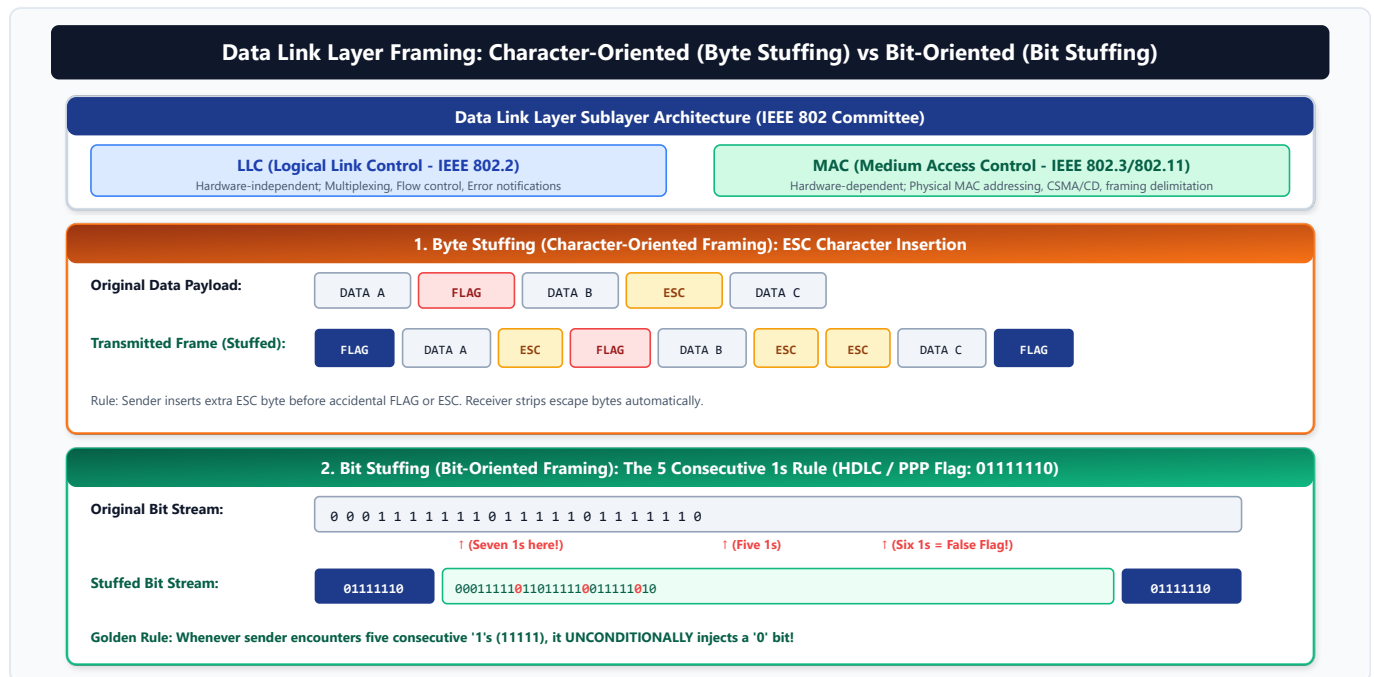


Module 01: Data Link Layer Foundations & Framing Techniques

AKTU BCS603 Core Topic: LLC vs MAC sublayers, Character Count vulnerability, Byte Stuffing escape sequences, and Bit Stuffing 5-ones rule.



1. Comparative Framing Strategies

Framing Technique	Demarcation Mechanism	Vulnerability / Overhead	Standard Protocol
Character Count	Header integer stores frame length	Single bit count error desynchronizes all subsequent frames	Early framing experiments (Obsolete)
Byte Stuffing	Flag bytes with ESC byte prefix	Tied to 8-bit character boundaries; high byte overhead	BISYNC, PPP (Byte mode)
Bit Stuffing	Flag pattern 01111110 with 0-bit injection after 5 ones	Negligible bit overhead (~1 bit per 32 bits on average); arbitrary data	HDLC, SDLC, PPP, Frame Relay

2. Golden Rule of Bit Stuffing

Sender Rule: Scan bit stream. Whenever five consecutive 1s (11111) are transmitted, **UNCONDITIONALLY** stuff a 0.

Receiver Rule: If five consecutive 1s are received:

- If next bit is 0: Destuff (strip) it!
- If next bit is 1 followed by 0: Valid **FLAG** detected!
- If next bit is 1 followed by 1: Error / Link Abort signal!